Faculty of Mathematics
and Computer Science

Knowledge Discovery (KD)

Steel Industry Knowledge Discovery on Continuous Casting Machine Dataset

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Abstract

This research addresses **predictive maintenance and process optimization in steel manufacturing**, focusing on the continuous casting stage. A **Continuous Casting Machine** (CCM) transforms molten steel into semi finished billets. A key component is the water cooled copper mold **sleeve**, which initiates solidification. Due to high thermal stress and friction, the sleeve degrades over time, making its monitoring important for reducing downtime and maintenance costs.

The main objective is to analyse the **Remaining Useful Life** (RUL) of mold sleeves using industrial data. A preprocessing pipeline was applied to correct this issue by unifying time and date into a single timeline and ordering records by sleeve identifier. A new RUL definition was introduced based on total processed steel, ensuring a monotonic and physically consistent measure.

For analysis, numerical variables were discretized into categorical levels (*Low*, *Medium*, *High*). The study combines machine learning and knowledge discovery methods. Supervised models, including **Random Forest Regressor** (RFR) and **Random Forest Classifier** (RFC), were used for baseline prediction and feature importance. Unsupervised methods, including DBSCAN, PCA, and **Isolation Forest** (IF), were applied for structure discovery and anomaly detection.

The main knowledge discovery method is **Formal Concept Analysis** (FCA), which organizes data into concept lattices representing relationships between process variables and equipment states. Association rule mining was also applied to extract logical dependencies. The analysis was supported by scaling methods implemented using the **ToscanaJ Suite**.

Advanced extensions include **Triadic Formal Concept Analysis** (3FCA), which introduces operating conditions as a third dimension, enabling analysis of context-dependent relationships. **Temporal Formal Concept Analysis** (TCA) incorporates time to model state transitions and reconstruct lifecycle trajectories.

Overall, combining machine learning with formal concept analysis enables structured interpretation of industrial data and improves understanding of mold sleeve degradation.

Keywords: (KD) Knowledge Discovery; (KDD) Knowledge Discovery in Databases; (AI) Artificial Intelligence; (ML) Machine Learning; (SL) Supervised Learning; (UL) Unsupervised Learning; (DA) Data Analysis; (CCM) Continuous Casting Machine; (SI) Steel Industry; (SE) Software Engineer; (DM) Data Mining; (FCA) Formal Concept Analysis; (TCA) Temporal Concept Analysis; (RUL) Remaining Useful Life; (RFR) Random Forest Regressor; (RFC) Random Forest Classifier; (DBSCAN) Density-Based Spatial Clustering of Applications with Noise; (PCA) Principal Component Analysis; (IF) Isolation Forest; (AE) Attribute Exploration; (3FCA) Triadic Formal Concept Analysis; (CTS) Conceptual Time System

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1. Knowledge Discovery

1.1. Knowledge Discovery

First of all, lets define every field approached within this document. **Knowledge Discovery** (Knowledge Discovery (KD)) is the process of extracting general, meaningful patterns from data rather than analysing isolated records [2]. Its objective is to transform raw data into structured and explainable knowledge that captures what is consistently true in the dataset.

KD focuses on identifying:

- **Patterns**: regular structures in data;
- **General rules**: valid beyond individual cases);
- **Stable relationships**: robust dependencies between variables.

1.2. Formal Concept Analysis

Formal Concept Analysis (FCA) is a mathematical framework for discovering structured knowledge in data [3]. It represents information as a set of **formal concepts**, each composed of:

- **Extent**: a set of objects;
- **Intent**: a set of shared attributes.

These concepts are organized into a **concept lattice**, which orders knowledge from general to specific concepts. FCA is useful in KD because it produces:

- Interpretable concept hierarchies;
- Logical implication rules (*if A then B*);
- Reduced and non-redundant knowledge representations.

Overall, FCA transforms raw data into structured conceptual systems, enabling the extraction of exact dependencies between attributes and supporting explainable knowledge discovery.

1.3. ToscanaJ and Elba

ToscanaJ is a conceptual data visualization tool used to display formal conceptual structures, such as concept lattices, derived from relational databases or in-memory data [1]. It provides a graphical interface for exploring conceptual schemas and supports both simple and nested lattice diagrams. Key capabilities include:

- Visualization of concept lattices using line diagrams;
- Interactive node selection and filtering of object sets;
- Visual encoding of data;
- Export of diagrams in formats such as SVG, PNG and JPEG;
- Integration with database views and external web-based attribute viewers.

Elba is a conceptual system editor used to construct formal contexts and conceptual schemas directly on top of relational databases. It supports database-aware modeling, allowing users to define scales, attributes and conceptual structures while interacting with underlying data sources. Key features include:

- Database connection support (JDBC, ODBC, MS Access);
- Scale generation tools for constructing conceptual attributes,
- Diagram editing and layout management based on structured grid systems;
- XML-based schema editing and export capabilities;

- SQL export for database integration and migration;
- Support for generating formal contexts and conceptual diagrams.

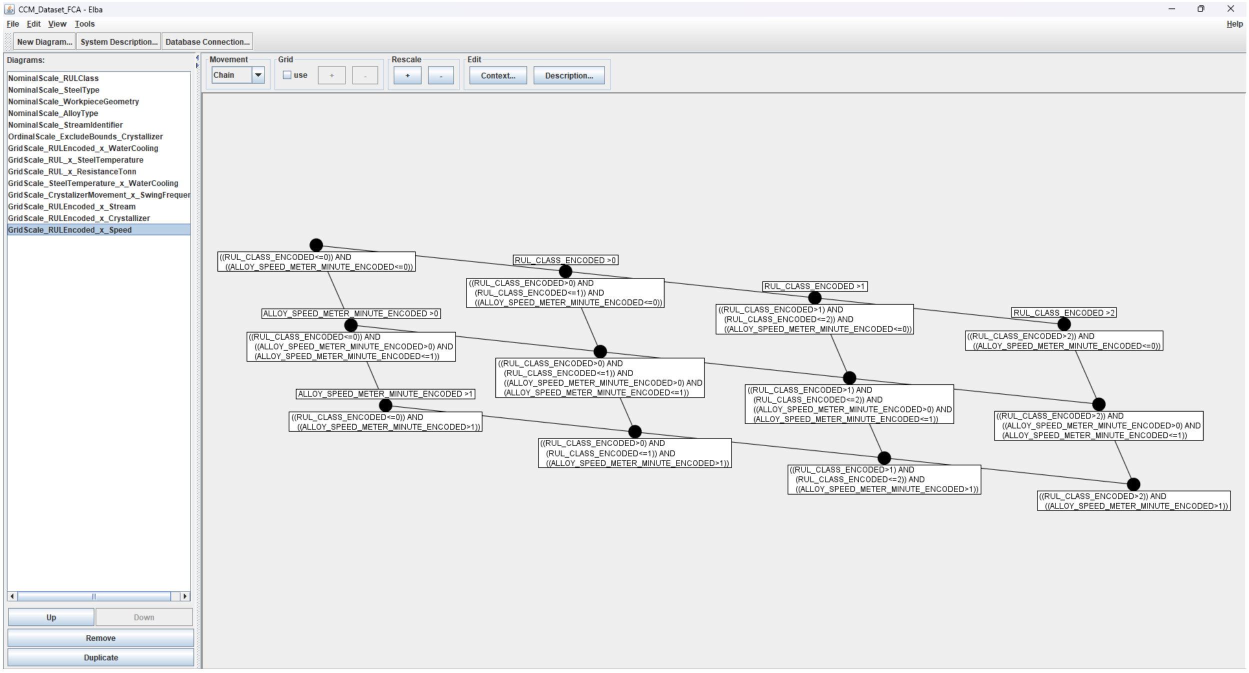

Fig. 1: **Elba** Application Sample

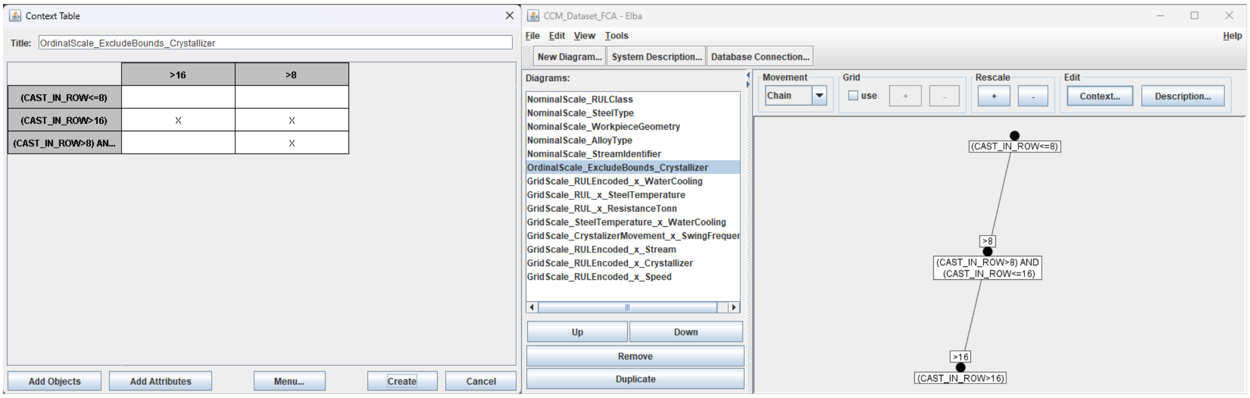

Fig. 2: **Elba** Application Sample

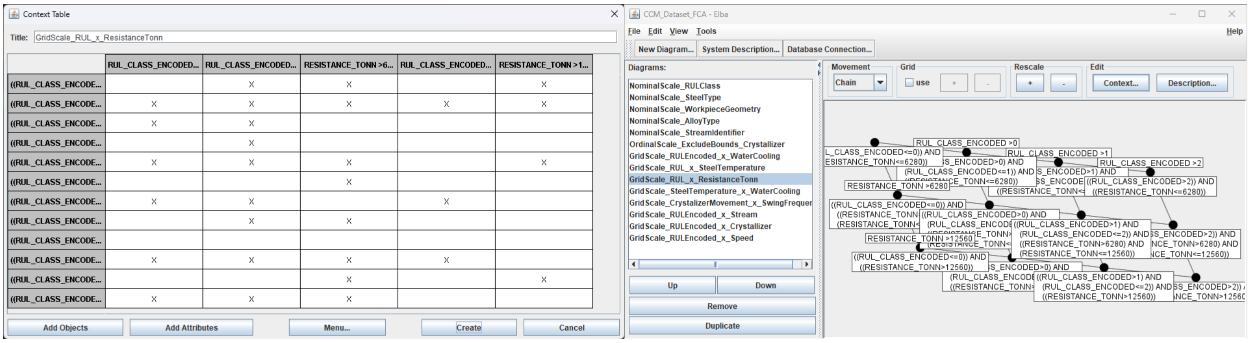

Fig. 3: **Elba** Application Sample

1.4. Conceptual Scales for FCA Analysis

The selected database for this analysis part is the preprocessed one. To enable FCA over the industrial dataset, it is necessary to define a set of **nominal**, **ordinal and grid based conceptual scales**. These scales discretize process variables into interpretable semantic categories (0 as Low, 1 as Medium and 2 as High for encoded variables; 0 as Critical, 1 as Low, 2 as Medium and 3 as Healty for Remaining Useful Life (RUL) class encoding), enabling the extraction of meaningful implication rules and concept lattices.

Scale Name	**Type**	**Values / Structure**
NominalScale_RUL_Class	Nominal	0 as Critical, 1 as Low, 2 as medium, 3 as Healthy
NominalScale_SteelType	Nominal	All processed steel types
NominalScale_WorkpieceGeometry	Nominal	150x150 or 180x180
NominalScale_AlloyType	Nominal	Open or closed
NominalScale_StreamIdentifier	Nominal	Identifier from 1 to 6
OrdinalScale_ExcludedBounds_CastInRow	Ordinal	<= 8 / > 8 and <= 16 / > 16 (or 1-8, 9-16, 17-24)
GridScale_RULEncoded_x_WaterCooling	Grid	RUL Class Encoded × Water Cooling Efficiency
GridScale_RUL_x_SteelTemperature	Grid	RUL Class Encoded × Steel Temperature Thermal Stress
GridScale_RULEncoded_x_ResistanceTonn	Grid	RUL Class Encoded × Resistance
GridScale_SteelTemperature_x_WaterCooling	Grid	Steel Temperature Thermal Stress × Water Cooling Efficiency
GridScale_RULEncoded_x_Stream	Grid	RUL Class Encoded × Stream Identifier
GridScale_RULEncoded_x_Crystallizer	Grid	RUL Class Encoded × Cristalyzer Identifier (1-8, 9-16, 17-24)
GridScale_RULEncoded_x_Speed	Grid	RUL Class Encoded × Alloy Speed Meter per Minute

Table 1: Formal Concept Analysis scales constructed in Elba

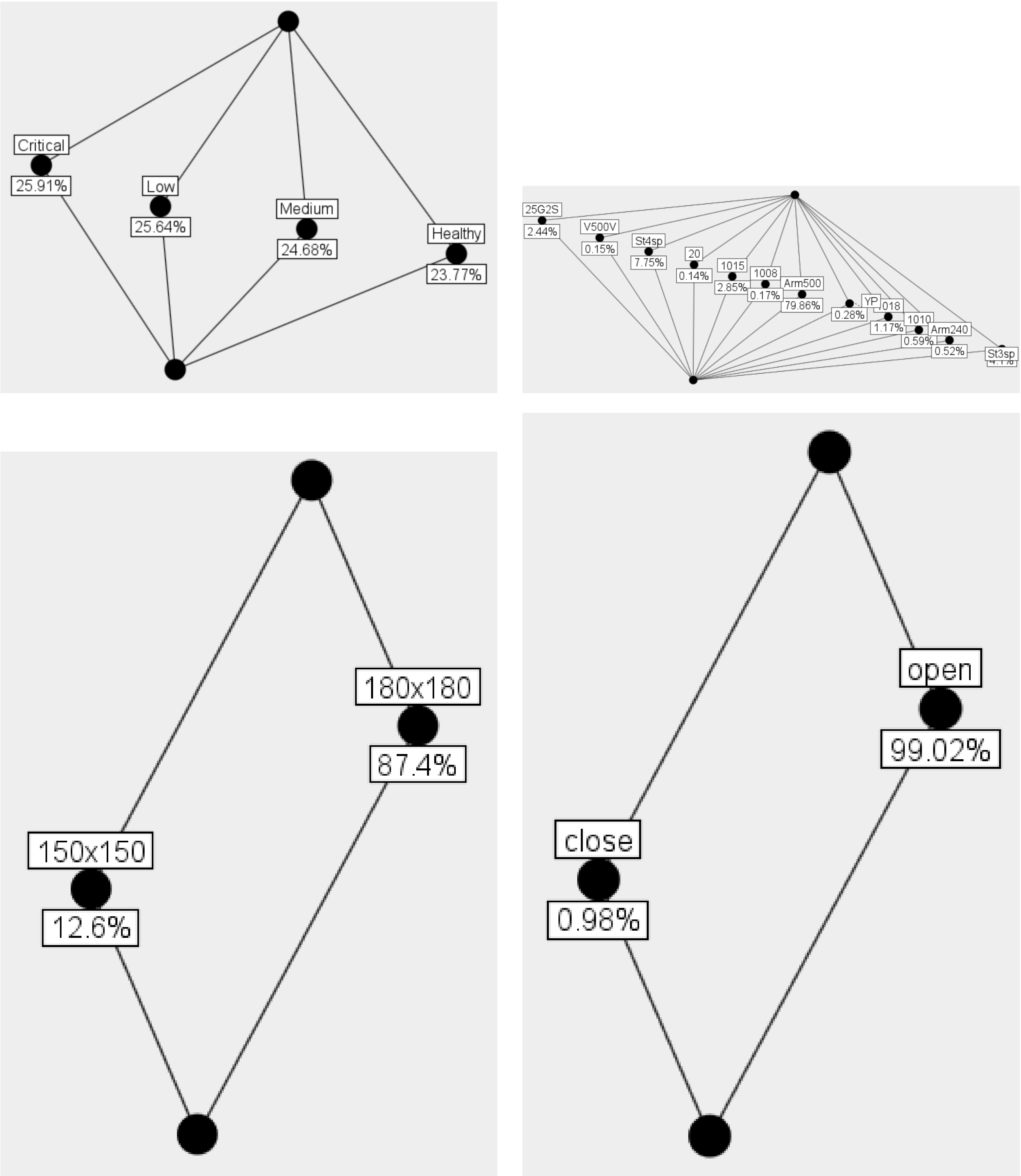

Fig. 4: ToscanaJ FCA Nominal Scales Representation

The previous **nominal scales** are a basic chart used in Formal Concept Analysis (FCA). They separate data into flat unconnected categories and act like a simple summary that shows the percentage of items in each group. They do not provide particular hidden gems for knowledge discovery.

Here is what the four previous provided nominal scales expose:

- **Alloy Type Distribution**: This scale shows that the casting process almost exclusively uses the "open" alloy type (99.02%). The "close" type is very rare (0.98%);
- **RUL Class Distribution:**: This scale shows a perfectly balanced dataset. The four categories (Critical, Low, Medium and Healthy) are almost equal in size, each holding roughly 24% to 26% of the data;
- **Steel Type Distribution**: This scale reveals a heavy preference for one specific material. The "**Arm500**" steel type makes up 79.86% of the records. The remaining data is spread across the other minor steel types;
- **Workpiece Geometry Distribution**: This scale tracks the physical shape of the cast steel. The larger "180×180" size is the most common at 87.4%, while the smaller "150×150" size makes up 12.6%.

More interesting results were obtained from the **grid scales**, which combine multiple attributes into a single analysis. These scales allow the exploration of interactions between process conditions and RUL classes, helping to identify patterns and relationships that are not visible when each attribute is analysed independently. As a result, the **grid scales provided a richer source of knowledge discovery** for the continuous casting process.

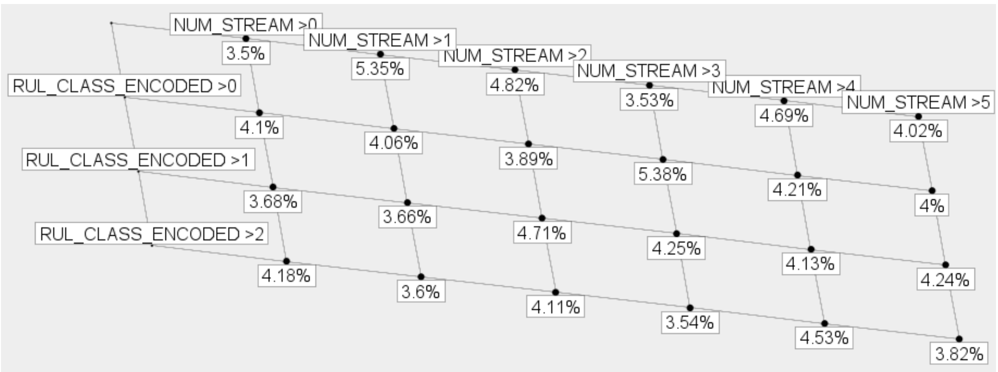

Fig. 5: ToscanaJ FCA Grid Scale Representation, RUL × Stream

In this grid scale, the columns represent the ordinal scaling of the *NUM_STREAM* attribute, while the rows represent the ordinal scaling of the *RUL_CLASS_ENCODED* attribute (row 1 represents value 0 meaning Critical, row 2 represents "> 0 *and* <= 1" as Low, "> 1 *and* <= 2" as Medium and "> 2" as Healty). The intersection at each cell contains the percentage of objects distribution belonging to that specific combined concept node. Values are almost equally spread trough nodes, varying between 3-5%.

RUL / NUM_STREAM	> 0 & <= 1	> 1 & <= 2	> 2 & <= 3	> 3 & <= 4	> 4 & <= 5	> 5
RUL <= 0	3.50%	5.35%	4.82%	3.53%	4.69%	4.02%
RUL > 0 & <= 1	4.10%	4.06%	3.89%	5.38%	4.21%	4.00%
RUL > 1 & <= 2	3.68%	3.66%	4.71%	4.25%	4.13%	4.24%
RUL > 2	4.18%	3.60%	4.11%	3.54%	4.53%	3.82%

Table 2: Grid scale interaction between NUM_STREAM and RUL_Class distribution

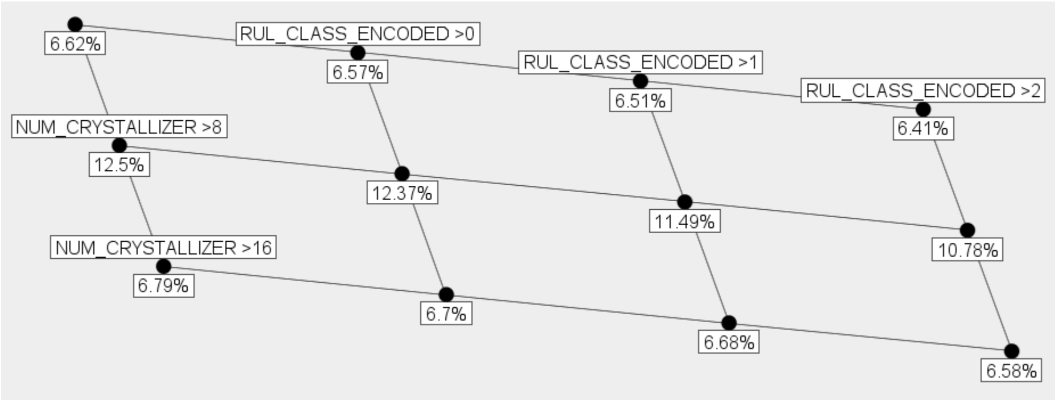

Fig. 6: ToscanaJ FCA Grid Scale Representation, RUL × Crystallizer

The grid scale analysis shows a relationship between the crystallizer used in the casting process and the RUL of the sleeve. The crystallizers is divided into three identifier groups for an easier visualization (1–8, 9–16 and 17+). The first and third groups have a balanced distribution of RUL classes, indicating a normal and stable degradation process.

The second group of crystallizers (9–16) shows a different pattern. This group processes a larger share of the production data and contains a higher proportion of Critical RUL cases compared to Healthy cases (column 3). This result suggests that crystallizers 9–16 may experience higher stress or less favourable operating conditions, leading to faster sleeve degradation. Therefore, this group represents a potential bottleneck in the production process and should be investigated to improve equipment lifetime and reduce maintenance costs.

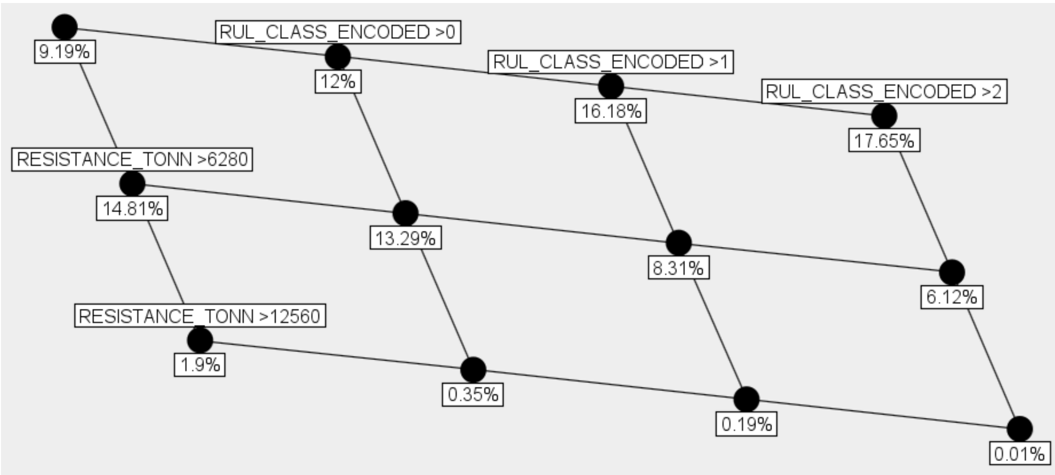

Fig. 7: ToscanaJ FCA Grid Scale Representation, RUL × Stream

The previous grid scale analyses the relationship between the amount of processed steel tonnage and the RUL of the sleeve. Sleeves with low processed tonnage are mostly found in the Healthy RUL class, indicating that the equipment performs well during the early stages of its life cycle.

As the processed tonnage increases, the distribution shifts toward the Critical and Low RUL classes. This indicates that sleeve degradation becomes more pronounced after a certain usage level. The middle tonnage range contains the largest proportion of Critical sleeves, showing that wear increases significantly during the mid life stage of the equipment.

The analysis also shows that **very few sleeves remain in operation after processing more than 12,560 tons of steel**. Among these cases, almost none belong to the Healthy class. This suggests that most sleeves reach a critical condition and are replaced.

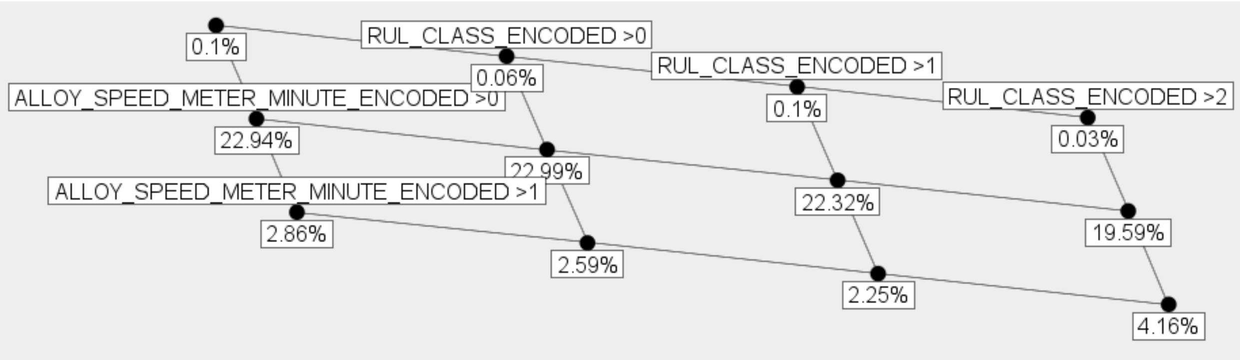

Fig. 8: ToscanaJ FCA Grid Scale Representation, RUL × Crystallizer

The previous grid scale shows the relationship between casting speed and RUL. Most production data belongs to the medium speed category, indicating that this is the normal operating speed of the casting process. In this range, the distribution contains slightly more Critical sleeves than Healthy sleeves, suggesting that equipment wear increases during regular operation.

The high-speed category shows a larger proportion of Healthy sleeves and a smaller proportion of Critical sleeves, suggesting that high casting speeds are probably used when the sleeve is in good condition. As the sleeve degrades and moves toward lower RUL, the casting speed is reduced to maintain process stability and product quality.

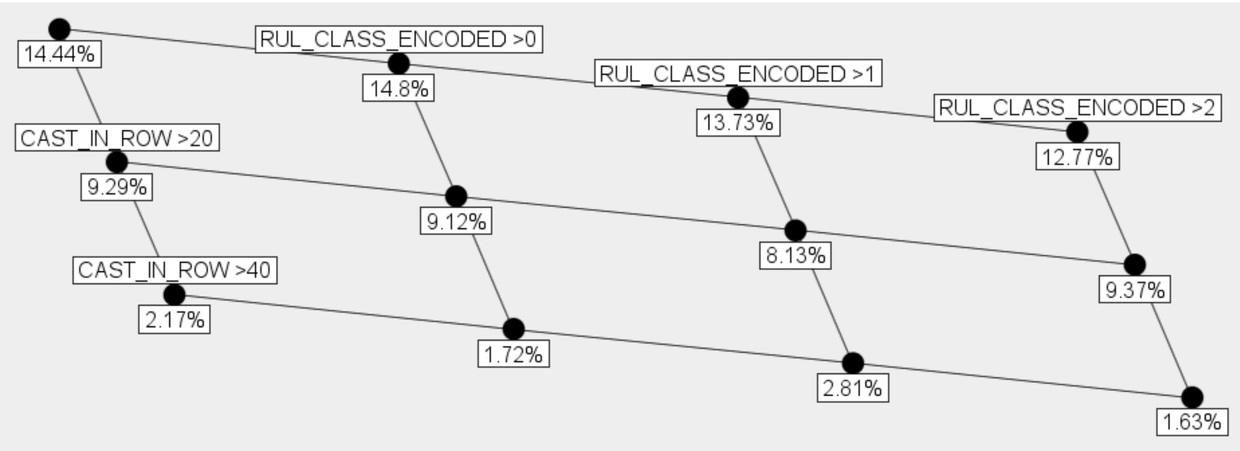

Fig. 9: ToscanaJ FCA Grid Scale Representation, RUL × Cast in row

The grid scale analysis shows the relationship between sleeve health RUL class and the length of continuous casting sequences (CAST_IN_ROW). As observed, most production happens in short sequences (0–20 casts), while fewer cases occur in medium (20+) and long (40+) sequences. In short sequences, the data shows more Critical sleeves than Healthy ones, **indicating higher wear during frequent stops and restarts**.

When the casting process runs for longer continuous sequences, the wear becomes more balanced between RUL classes, with similar proportions of Critical and Healthy sleeves. This suggests that **stable long production runs reduce degradation**.

Next, the dataset was further explored by creating three inter-ordinal scales in Elba and analyzing them in ToscanaJ through node navigation.

The three scales are defined as:

- ”**INTERORDINAL_STEEL_TEMPERATURE**”: an increasing inter-ordinal scale with inclusive bounds, built from the encoded binary concepts Low, Medium, and High. The values were extracted from the dataset minimum and maximum temperatures (567–1667 °C), using five thresholds: 750 °C, 933 °C, 1116 °C, 1300 °C, and 1483 °C;
- ”**INTERORDINAL_STEEL_WEIGHT_TONN**”: an increasing inter-ordinal scale built in the same way as the previous one. The values were extracted from the dataset minimum and maximum steel weights (110–188 t), using five thresholds: 123, 136, 154, 162, and 175 tons;
- ”**INTERORDINAL_WATER_CONSUMPTION**”: an increasing inter-ordinal scale built from the dataset minimum and maximum water consumption values (1255–2155 l/min), using five thresholds: 1405, 1555, 1705, 1855, and 2005 liters per minute.

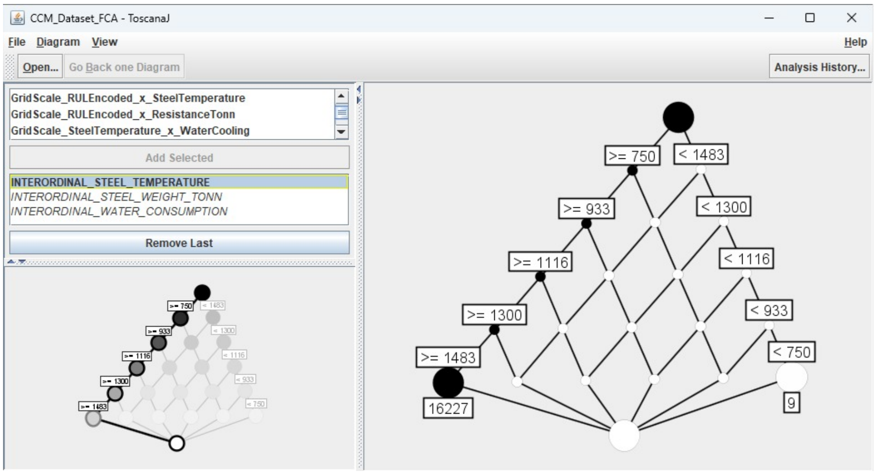

Fig. 10: ToscanaJ INTERORDINAL_STEEL_TEMPERATURE Scale Navigation

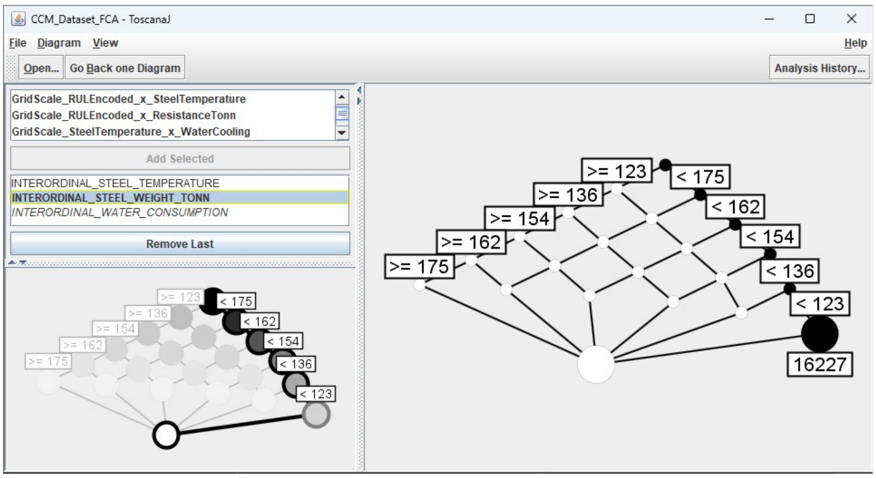

Fig. 11: ToscanaJ INTERORDINAL_STEEL_WEIGHT_TONN Scale Navigation

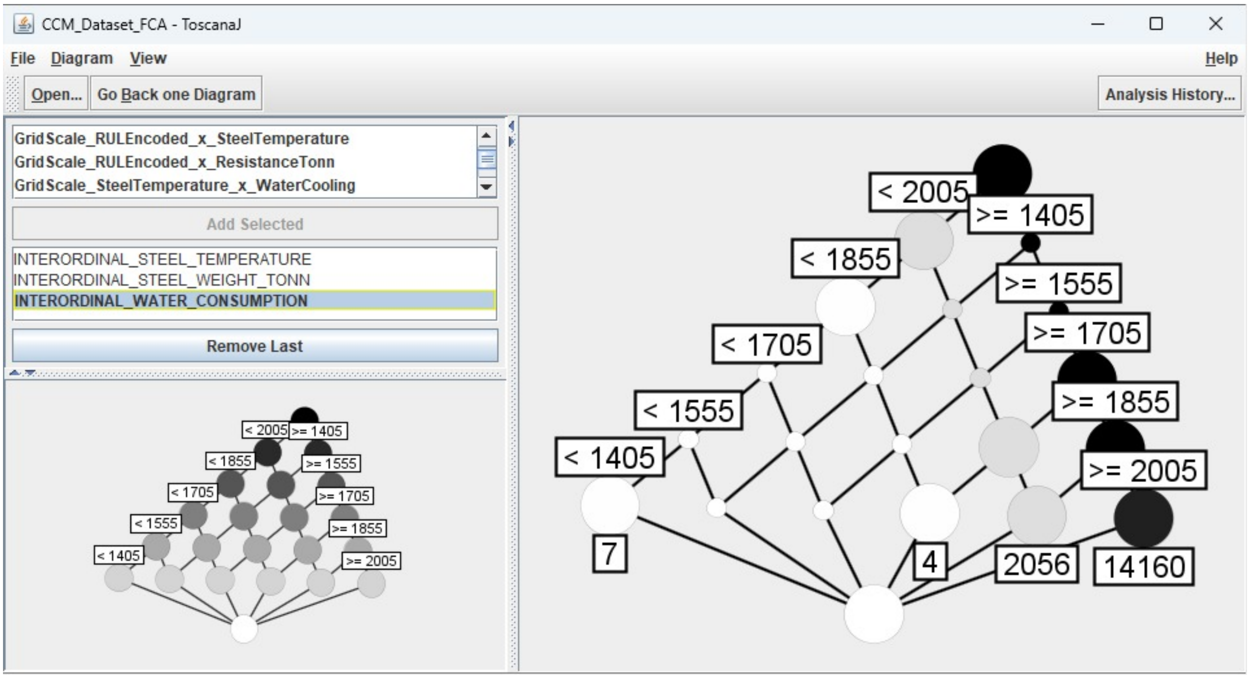

Fig. 12: ToscanaJ INTERORDINAL_WATER_CONSUMPTION Scale Navigation

The navigation of the first scale shows that most dataset records have a steel temperature greater than or equal to 1483 °C. Only 9 records (less than 0.1% of the dataset) have a temperature below 750 °C.

By navigating to the node with the highest number of records (≥ 1483), the system displays the distribution of steel production weight. Most of these records belong to the lower weight interval, with values below 123 tons.

Further navigation to the third scale shows the water consumption for these records. In this case, the distribution is divided into two main groups: 14 160 records with water consumption greater than or equal to 2005 liters per minute, and 2 056 records with values between $1855 \leq n < 2005$ liters per minute.

This **analysis describes the normal operating conditions of the Continuous Casting Machine**. Records that differ from these common patterns can be considered anomalies or non-standard operating behaviours.

2. Attribute Exploration

2.1. Attribute Exploration

First of all, lets define what **Attribute Exploration** is. Attribute Exploration (AE) is a method from Formal Concept Analysis (FCA) used to discover all valid attribute dependencies in a dataset. It starts from a formal context, which is a table describing which objects have which attributes. The goal is to identify implication rules of the form (A *to* B), meaning that whenever an object has all attributes in (A), it must also have all attributes in (B) [5].

The method works by systematically generating candidate implications and checking whether they are valid in the dataset. If a candidate implication is correct, it is added to a growing set of rules, else if it is not correct, a counter example is required. This counterexample is then added to the dataset to refine the analysis.

A key idea in AE is that the process continues until no new valid implications can be found. The resulting set of implications forms a complete and non redundant description of the relationships between attributes. The method is strongly connected to closure systems and logical reasoning.

2.2. Attribute Exploration FCA Setup

For this research part, the "**ConExp**" application was used. Tools name stands for **Concept Explorer** and it is a tool widely used for studying and applying Formal Concept Analysis, offering context editing, lattice generation, implications and association rules, and finally **attribute exploration** [6].

The dataset was preprocessed to generate a **binary formal context** compatible with the tool. A python script was used to selects only the binary (`_BIN`) attributes (containing values such as Low/Medium/High, output from the dataset pre-processing step), removes constant attributes that do not provide useful information and eliminates duplicate objects to reduce redundancy. Finally, **nominal conceptual scaling** is applied by converting each attribute into a binary representation using indicator variables and the resulting formal context is exported as a CSV file for use in the attribute exploration process.

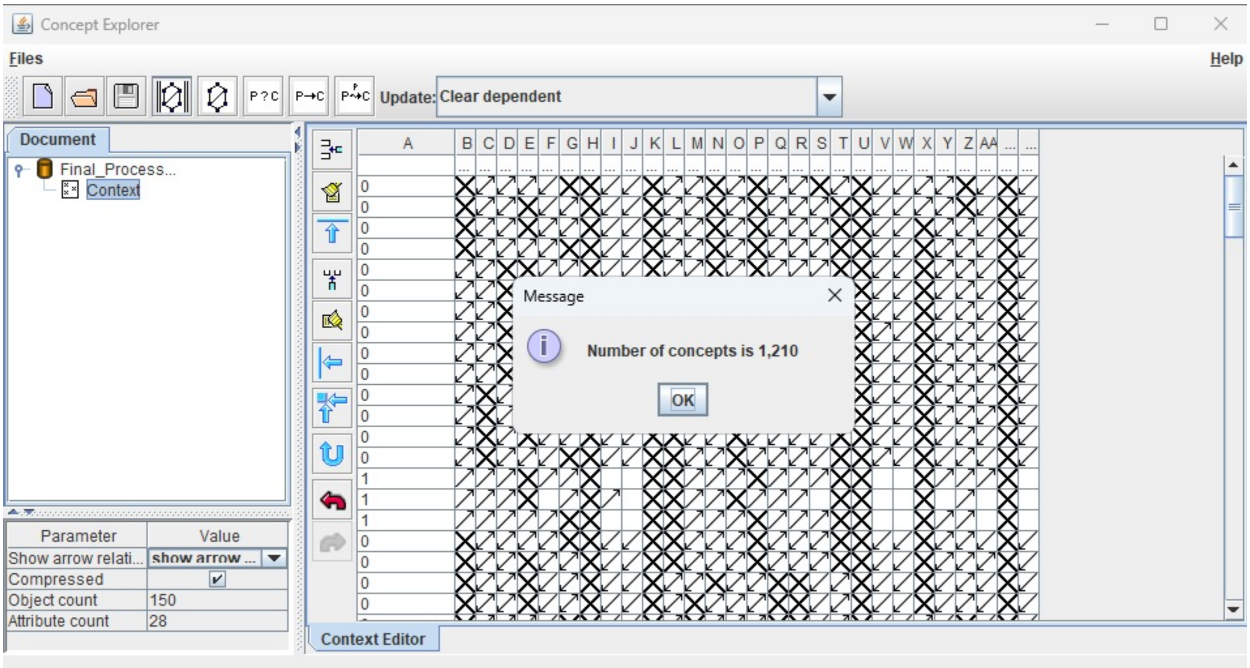

Fig. 13: ConExp Attribute Exploration

After loading the dataset into the tool, the cleaning functions were used to remove non relevant attributes and empty rows. Once the formal context was prepared, the "*Start Attribute Exploration*" function was executed. During this process, the tool generated a series of implication rules that required validation. Each rule was evaluated using domain knowledge and practical experience in the steel manufacturing industry. After the Attribute Exploration step was complete, the "*Calculate Duquenne-Guigues set of implication*" feature was executed and finally the "*Calculate association rules*".

The attribute exploration process produced more than 500 implication cases, even after the initial preprocessing steps. For this reason, only 10 representative cases were selected for presentation. Five of these implications were accepted, while the other five were rejected.

- **Question 1**: Does `swing_frequency_amount_minute_BIN=High` ⇒ `steel_weight_tonn_BIN=Medium` ∧ `resistancetonn_BIN=High` ∧ `temperature_measurement1Celsius_BIN=High`?
  Decision: Accepted.
  Explanation: The dataset shows this implication with 100% confidence. High swing frequency is always associated with medium steel weight, high resistance and high temperature measurement in the observed production data;

- **Question 2**: Does `crystallizer_movementmm_BIN=Low` ⇒ `steel_weight_tonn_BIN=Medium` ∧ `resistancetonn_BIN=High` ∧ `alloy_speed_meter_minute_BIN=Medium` ∧ `temperature_measurement1Celsius_BIN=High`?
  Decision: Accepted.
  Explanation: The implication has 100% confidence. Every production record with low crystallizer movement also contains these operating conditions;
- **Question 3**: Does `swing_frequency_amount_minute_BIN=Medium` ∧ `alloy_speed_meter_minute_BIN=Low` ⇒ `temperature_measurement1Celsius_BIN=High`?
  Decision: Accepted.
  Explanation: The implication is supported by all corresponding objects in the dataset. Temperature measurement is always high when these two process conditions occur together;
- **Question 4**: Does `temperature_measurement1Celsius_BIN=High` ⇒ `resistancetonn_BIN=High`?
  Decision: Accepted.
  Explanation: The implication has approximately 99% confidence. Almost every object with a high temperature measurement also has high resistance, making this a very strong dependency;
- **Question 5**: Does `alloy_speed_meter_minute_BIN=Medium` ⇒ `resistancetonn_BIN=High`?
  Decision: Accepted.
  Explanation: The implication has approximately 99% confidence. Medium alloy speed is strongly related to high resistance in the production process;
- **Question 6**: Does `FCA_BIN=Critical` ∧ `resistancetonn_BIN=High` ∧ `temperature_measurement1Celsius_BIN=High` ⇒ `alloy_speed_meter_minute_BIN=Medium`?
  Decision: Rejected.
  Counter example explanation: A production case can be introduced where the equipment is in the Critical state and has high resistance and high temperature, but the alloy speed is High instead of Medium;
- **Question 7**: Does `FCA_BIN=Medium` ∧ `resistancetonn_BIN=High` ∧ `alloy_speed_meter_minute_BIN=Medium` ⇒ `steel_weight_tonn_BIN=Medium`?
  Decision: Rejected.
  Counter example explanation: A production record may satisfy the premise while having High steel weight instead of Medium. This would violate the implication;
- **Question 8**: Does `FCA_BIN=Healthy` ∧ `steel_weight_tonn_BIN=Medium` ∧ `resistancetonn_BIN=High` ∧ `alloy_speed_meter_minute_BIN=Low` ⇒ `temperature_measurement1Celsius_BIN=High`?
  Decision: Rejected.
  Counter example explanation: A healthy production sequence can be constructed where temperature measurement is Medium instead of High while the remaining premise attributes are satisfied;
- **Question 9**: Does `FCA_BIN=Low` ∧ `resistancetonn_BIN=High` ∧ `water_consumption_BIN=Medium` ∧ `temperature_measurement1Celsius_BIN=High` ⇒ `steel_weight_tonn_BIN=Medium`?
  Decision: Rejected.
  Counter example explanation: Another production case may satisfy all premise conditions while steel weight belongs to the High category instead of Medium;
- **Question 10**: Does `FCA_BIN=Healthy` ∧ `FCA_BIN=Medium` ∧ `steel_weight_tonn_BIN=Medium` ∧ `resistancetonn_BIN=High` ∧ `crystallizer_movementmm_BIN=High` ∧ `alloy_speed_meter_minute_BIN=Medium` ∧ `water_consumption_BIN=Medium` ⇒ `temperature_measurement1Celsius_BIN=High`?
  Decision: Rejected.
  Counter example explanation: The implication has only about 80% confidence. A production object can satisfy the premise while temperature measurement is Medium, showing that the implication is not universally valid.

2.3. Attribute Exploration Interpretation

The attribute exploration process identified several dependencies between the monitored process variables. The accepted implications show that **high temperature, high resistance, medium steel weight and specific operating parameters** frequently occur together during normal production. These rules represent stable relationships that are consistently observed in the dataset.

The rejected implications demonstrate that some relationships are not universally valid. By introducing counterexamples, it was shown that different operating conditions can produce alternative outcomes even when most premise attributes are satisfied. This indicates that these dependencies cannot be considered general rules of the production process.

Overall, the attribute exploration process confirmed several strong process relationships while also identifying assumptions that cannot be generalized. The resulting implication base provides a compact representation of the knowledge extracted from the steel production dataset and can support further analysis of **process behaviour and anomaly detection**.

3. Triadic Knowledge Discovery

3.1. Triadic Formal Concept Analysis

In previous section 1 normal FCA was performed on two dimensions. Triadic Formal Concept Analysis (3FCA) is an extension of standard FCA that works with data structured in three dimensions instead of two. While classical FCA connects **objects and attributes**, 3FCA adds a third component called **conditions** [7].

A triadic context is defined by three sets together with a relation that links all of them: **objects, attributes and conditions**. A relation entry means that an object has a specific attribute under a specific condition. This structure can be represented as a three dimensional data cube or as a collection of two dimensional tables, each corresponding to one condition.

In 3FCA the main analytical unit or relation between sets is the **triconcept**. Every object in the set has all attributes under all conditions in the group. Triconcepts describe stable patterns that hold across all three dimensions at the same time. This allows more detailed pattern discovery compared to classical FCA, especially in domains where behaviour changes across different environments or situations.

3.2. 3FCA on Continuous Casting Data

The original dataset contains one record for every casting operation together with the measured process variables (sensor data) and production information. Since **several consecutive casting operations belong to the same sleeve**, using every row directly as an object would produce many repeated patterns and highly redundant concepts.

To obtain a more meaningful triadic context, the data was first grouped by the `sleeve` identifier, becoming one object in the triadic context. For every sleeve, the most frequent (majority) value was computed for each categorical attribute and each discretized process variable, producing one representative description for every sleeve.

A Python preprocessing script was developed to automate this transformation. The script performs the following steps:

1. Groups all records having the same sleeve identifier;
2. Computes the majority value of every categorical attribute and discretized process variable;
3. Generates the triadic formal context (3FCA);
4. Exports the resulting context as a CSV file that can be imported into the **FCA Tools Bundle** ([4]).

The resulting triadic formal context is defined as follows:

- **Objects (G):** casting sleeves;
- **Attributes (M):** production descriptors such as steel type, alloy type, workpiece geometry, remaining useful life category (RUL_BIN), number of streams and number of crystallizers;
- **Conditions (B):** discretized process measurements including casting order, steel weight, steel temperature, resistance, swing frequency, crystallizer movement, alloy speed, water consumption, water temperature difference and temperature measurements.

Each tuple $(g, m, b) \in Y$ indicates that the sleeve g possesses the production attribute m under the operating condition b. The generated context was exported as a CSV file with the following structure:

Object	Attribute	Condition
30011717	steel_type=Arm500	steel_weighttonn_BIN=High
30011717	steel_type=Arm500	resistance_tonn_BIN=Low
30011717	alloy_type=open	alloy_speed_meter_minute_BIN=Medium
...	...	...

A	B	C
30012903	steel_type=Arm500	resistance_tonn_BIN=Low
30012903	steel_type=Arm500	steel_temperature_grab1Celsius_BIN=High
30012903	steel_type=Arm500	steel_weighttonn_BIN=High
30012903	steel_type=Arm500	swing_frequency_amount_minute_BIN=High
30012903	steel_type=Arm500	temperature_measurement1_Celsius_BIN=Low
30012903	steel_type=Arm500	temperature_measurement2_Celsius_BIN=Low
30012903	steel_type=Arm500	water_consumption_BIN=High
30012903	steel_type=Arm500	water_temperature_deltaCelsius_BIN=High
30012903	workpiece_slice_geometry=180x180	alloy_speed_meter_minute_BIN=Medium
30012903	workpiece_slice_geometry=180x180	crystallizer_movementmm_BIN=Low
30012903	workpiece_slice_geometry=180x180	resistance_tonn_BIN=Low
30012903	workpiece_slice_geometry=180x180	steel_temperature_grab1Celsius_BIN=High
30012903	workpiece_slice_geometry=180x180	steel_weighttonn_BIN=High
30012903	workpiece_slice_geometry=180x180	swing_frequency_amount_minute_BIN=High
30012903	workpiece_slice_geometry=180x180	temperature_measurement1_Celsius_BIN=Low
30012903	workpiece_slice_geometry=180x180	temperature_measurement2_Celsius_BIN=Low
30012903	workpiece_slice_geometry=180x180	water_consumption_BIN=High
30012903	workpiece_slice_geometry=180x180	water_temperature_deltaCelsius_BIN=High
30012904	RUL_BIN=Critical	alloy_speed_meter_minute_BIN=Medium
30012904	RUL_BIN=Critical	crystallizer_movementmm_BIN=Low
30012904	RUL_BIN=Critical	resistance_tonn_BIN=Low
30012904	RUL_BIN=Critical	steel_temperature_grab1Celsius_BIN=High
30012904	RUL_BIN=Critical	steel_weighttonn_BIN=High

Fig. 14: 3FCA preprocessed dataset visualization

This format is compatible with the **FCA Tools Bundle** and can be processed for triconcept extraction. A triconcept represents a group of casting sleeves that share the same production characteristics under the same operating conditions. The new triadic formal context was reduced to only 5 objects in order to reduce input size (sleeve identifiers *30012903, 30012904, 30012905, 30012907 and 30012909*).

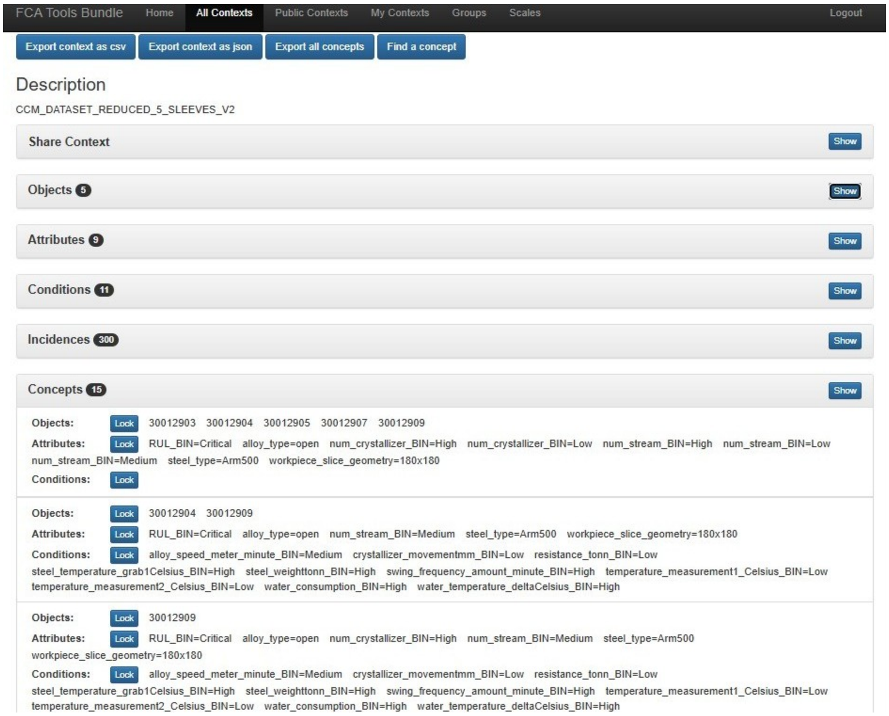

Fig. 15: FCA Tools Bundle Computed Concepts

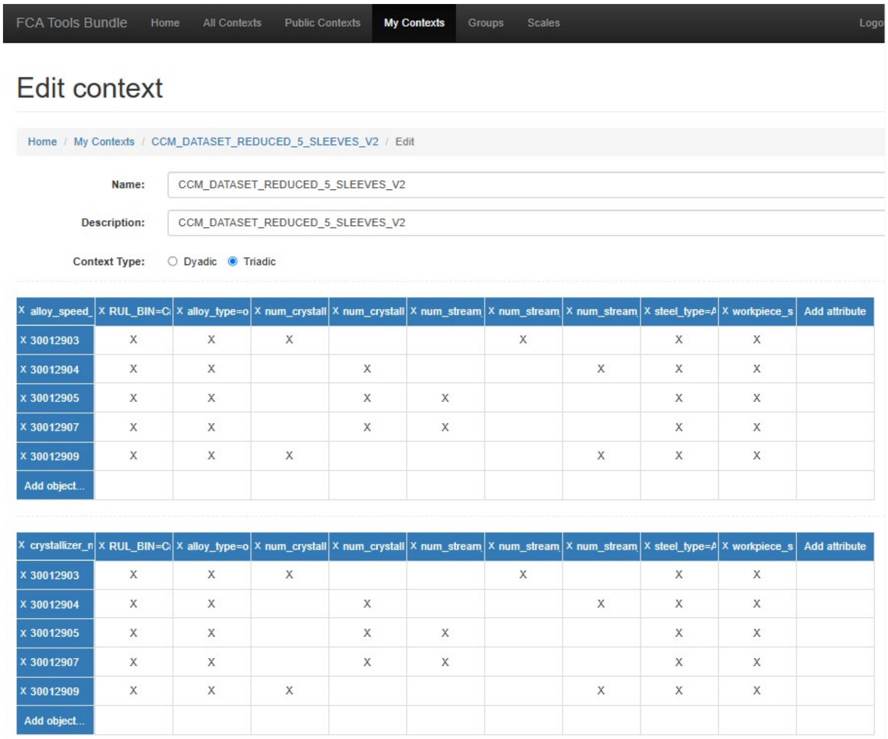

Fig. 16: FCA Tools Bundle Triadic Formal Context Visualization

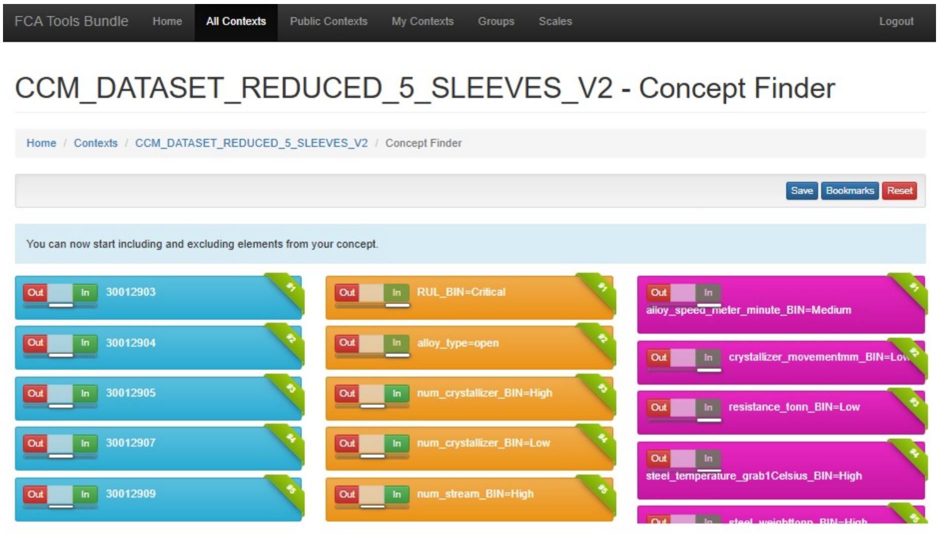

Fig. 17: FCA Tools Bundle Rule Filtering

The extracted triconcepts were used to generate conditional implication rules of the form $A \rightarrow B \mid C$ where:

- A is the antecedent set of attributes;
- B is the consequent set of attributes;
- C is the set of process conditions under which the implication holds.

Table 3: Found Triadic Association Rules over 192 discovered ones

Rule ($A \rightarrow B \mid C$)	**Support**	**Confidence**
{alloy_type=open} → {steel_type=Arm500} | {alloy_speed_meter_minute_BIN=Medium}	100%	100%
{RUL_BIN=Critical} → {alloy_type=open} | {steel_weighttonn_BIN=High}	100%	100%
{steel_type=Arm500} → {workpiece_slice_geometry=180x180} | {water_consumption_BIN=High}	100%	100%
{RUL_BIN=Critical} → {alloy_type=open} | {alloy_speed_meter_minute_BIN=Medium}	80%	100%
{RUL_BIN=Critical} → {steel_type=Arm500} | {swing_frequency_amount_minute_BIN=High}	80%	100%
{alloy_type=open} → {RUL_BIN=Critical} | {alloy_speed_meter_minute_BIN=Medium}	80%	80%
{steel_type=Arm500} → {RUL_BIN=Critical} | {alloy_speed_meter_minute_BIN=Medium}	80%	80%
{RUL_BIN=Critical} → {num_crystallizer_BIN=Low} | {alloy_speed_meter_minute_BIN=Medium}	60%	75%
{alloy_type=open} → {num_crystallizer_BIN=Low} | {steel_weighttonn_BIN=High}	60%	60%
{steel_type=Arm500} → {num_crystallizer_BIN=Low} | {water_consumption_BIN=High}	60%	60%

The extracted rules show several deterministic relationships with 100% confidence. These rules indicate that **some production attributes always appear together under specific operating conditions in the analysed dataset**. For example the steel type *Arm500*, the *open* alloy type and the *180×180* workpiece geometry frequently occur together.

Rules with 80% confidence indicate strong but not perfect relationships. In these cases, the implication is valid for most of the analysed sleeves, but there are some exceptions.

The rule with 75% confidence shows that the Critical RUL category is often associated with a low number of crystallizers when the alloy speed is medium. Finally, the rules with 60% confidence represent weaker relationships. These rules suggest possible associations, but they occur only in part of the analysed production data.

Overall, **the extracted rules indicate that the analysed dataset contains several stable operating patterns**. The high number of rules with perfect confidence also suggests that the dataset is relatively homogeneous after the aggregation by sleeve.

To better understand the structure of the triadic context, each condition was projected into a **dyadic context**. The dyadic context slice was created by fixing one condition from the triadic context and keeping only the *object-attribute relations* that satisfy that condition. A concept lattice was then generated for every slice, and the number of formal concepts was used as a measure of the structural complexity of that condition.

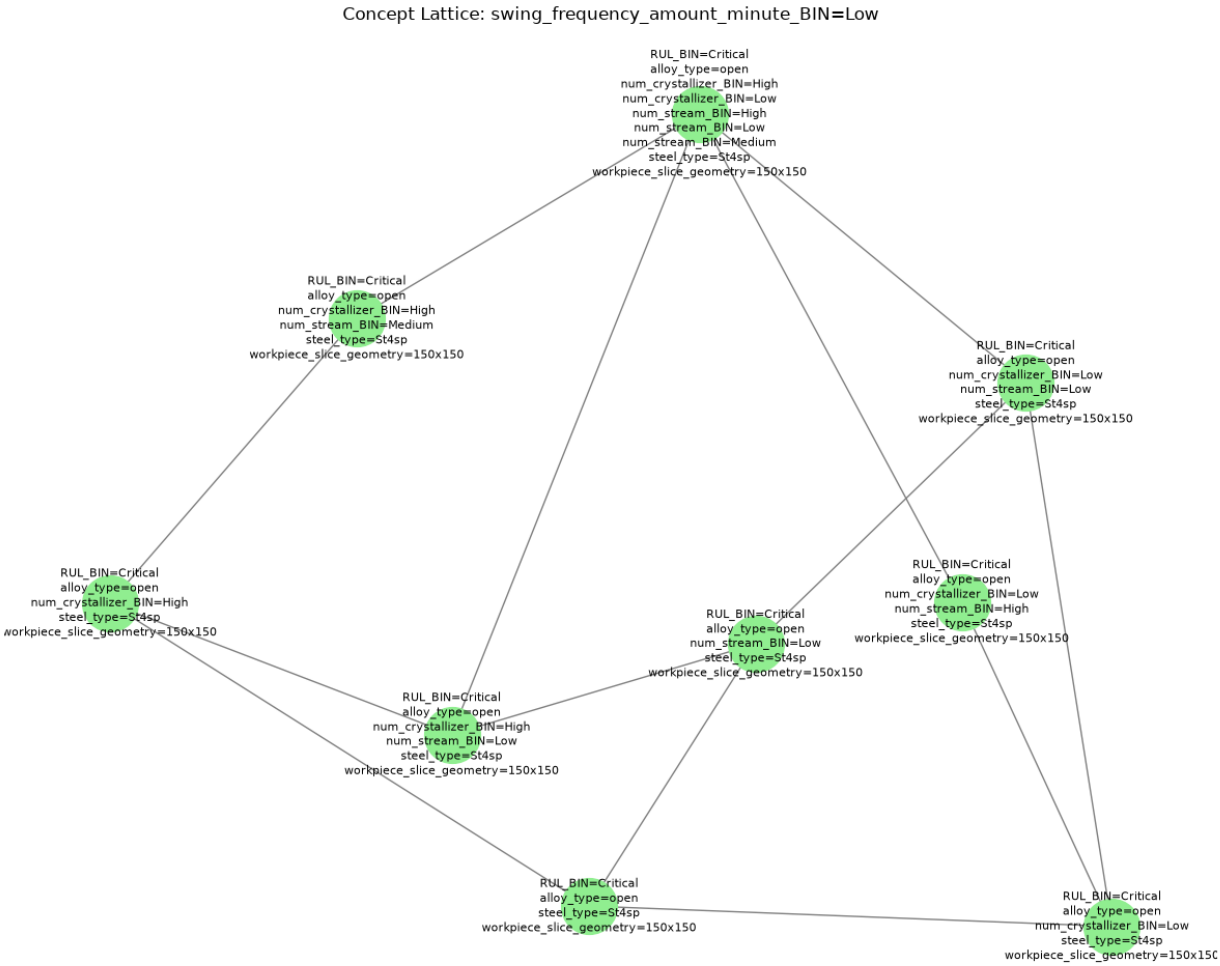

Fig. 18: Sample Concept Lattice Dyadic Context for the condition `swing_frequency_amount_minute_BIN=Low`

The labels attached to each node show the attributes that are common to the objects contained in that concept. The nodes located higher in the lattice represent more general concepts, while the nodes located lower represent more specific concepts with additional shared attributes. The lattice also shows that several attributes frequently occur together for this condition. For example, attributes such as `RUL_BIN=Critical`, `alloy_type=open` and `workpiece_slice_geometry=150x150` appear in many concepts. This indicates that these characteristics are strongly associated when the swing frequency belongs to the `Low` category.

The following table presents the number of objects, attributes and formal concepts for every condition, together with the ratio between the number of concepts and the number of objects.

Table 4: Per-condition dyadic context slices and their concept lattice sizes.

Condition	Objects	Attributes	Concepts	Concepts/Object
steel_temperature_grab1Celsius_BIN=High	86	16	98	1.14
temperature_measurement1_Celsius_BIN=Low	86	16	98	1.14
temperature_measurement2_Celsius_BIN=Low	86	16	98	1.14
water_consumption_BIN=High	86	16	98	1.14
water_temperature_deltaCelsius_BIN=High	86	16	98	1.14
resistance_tonn_BIN=Low	84	15	86	1.02
steel_weighttonn_BIN=High	80	15	86	1.07
alloy_speed_meter_minute_BIN=Medium	76	13	54	0.71
swing_frequency_amount_minute_BIN=High	66	13	49	0.74
crystallizer_movementmm_BIN=Low	71	12	42	0.59
swing_frequency_amount_minute_BIN=Medium	14	14	31	2.21
crystallizer_movementmm_BIN=High	12	13	30	2.50
alloy_speed_meter_minute_BIN=High	10	11	19	1.90
steel_weighttonn_BIN=Medium	6	9	15	2.50
swing_frequency_amount_minute_BIN=Low	6	9	9	1.50
crystallizer_movementmm_BIN=Medium	3	8	4	1.33
resistance_tonn_BIN=High	2	7	4	2.00

The largest concept lattices were obtained for the conditions `steel_temperature_grab1Celsius_BIN=High`, `temperature_measurement1_Celsius_BIN=Low`, `temperature_measurement2_Celsius_BIN=Low`, `water_consumption_BIN=High`, and `water_temperature_deltaCelsius_BIN=High`. Each slice contains 98 formal concepts generated from 86 objects and 16 attributes, showing that these conditions are common while preserving many attribute combinations.

The conditions `resistance_tonn_BIN=Low` and `steel_weighttonn_BIN=High` also produced large lattices with 86 concepts, indicating a rich concept structure.

Some less frequent conditions have a high concepts-to-object ratio. For example, `crystallizer_movementmm_BIN=High` and `steel_weighttonn_BIN=Medium` both have a ratio of 2.50 concepts per object, showing greater diversity among the available objects.

The smallest lattices were obtained for `crystallizer_movementmm_BIN=Medium` and `resistance_tonn_BIN=High`, each containing only four concepts because these conditions appear only a few times in the dataset.

The results show that **common operating conditions generate larger concept lattices**, while **rare conditions produce smaller and simpler lattices**, indicating different levels of conceptual complexity in the triadic context.

3.3. Triadic Concept Analysis Interpretations

The 3FCA analysis **identified stable relationships between production attributes and operating conditions through triconcepts and conditional implication rules**. The extracted rules and concept lattices show that common operating conditions generate richer conceptual structures, while less frequent conditions produce simpler patterns. These results demonstrate that 3FCA is suitable for discovering stable process knowledge in the Continuous Casting Machine (CCM) dataset.

4. Temporal Knowledge Discovery

4.1. Temporal Concept Analysis

Temporal Concept Analysis (Temporal Concept Analysis (TCA)) is an extension of FCA that is used to **analyse how structured knowledge changes over time**. While FCA describes data in a static form using objects, attributes and formal concepts, **TCA adds a temporal dimension** [8]. This allows data to be observed at different time points, called time granules, where each time point is described by a formal context representing the state of the system at that moment.

In TCA, each time granule is mapped to a formal concept in a concept lattice, which represents the state of the system at that time. By connecting consecutive time points, transitions between concepts are formed, creating a state-transition structure. This structure shows how the system evolves from one conceptual state to another over time, making it possible to analyse changes in a systematic way.

A key idea in TCA is the life-track, which describes the evolution of an individual object across time by mapping each time point to a corresponding formal concept. This produces a trajectory through the concept lattice that can be used to study patterns such as stability, change or repetition.

Given the fact that time is divided into steps, a concept is "alive" if it exists during a specific time step. This means a set of objects shares an exact set of features at that time. The "life track" of a concept is the list of time steps where it appears. The "lifespan" is the total number of time steps the concept exists.

4.2. Temporal Concept Analysis in Continuous Casting Data

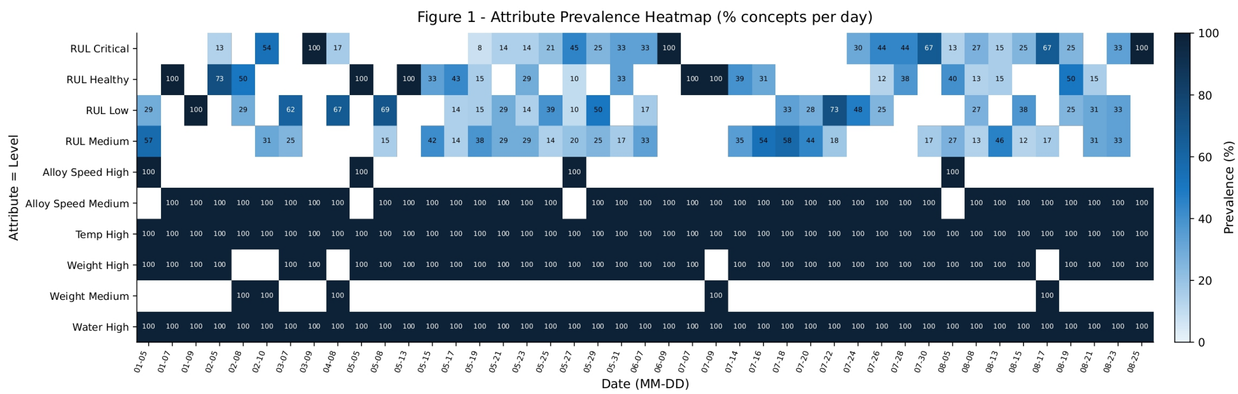

Fig. 19: TCA Attribute Prevalence Heatmap

The previous heatmap displays how often specific machine features appear over several days.

In the heatmap, **each vertical column on the x-axis is one time granule**. For example, the date "01-05" is one time granule. The date "01-07" is the next time granule. Machine data is grouped and analysed separately for each day.

In the heatmap, the numbers inside the boxes show the prevalence of a attribute. Prevalence is the percentage of formal concepts in that time granule that have the attribute. For example, on the date "02-10", the attribute "RUL Medium" has a value of 31. This means 31% of the formal concepts on that day included the "RUL Medium" attribute.

An **event part** is a attribute that defines a specific event or a forced change in the system. These features show sudden, full changes instead of slow changes. They show events like a machine setup change.

In the heatmap, features for "Alloy Speed" and "Weight" are event parts. They usually appear at 100% and then suddenly change to another level. For example, "Weight Medium" is at exactly 100% on specific days like "02-08", "02-10", "04-08", "07-09", and "08-17". On these same days, "Weight High" drops to 0%. This sudden change is an operational event. It shows a change in the steel production recipe for those days.

A **time part** is a attribute that depends on an event or changes slowly over time. These features react to time passing or the state of the system. They are rarely at a constant 100% or 0%.

In the heatmap, **features named RUL are time parts**. They show the physical wear of machine parts, a continuous process. For example, "RUL Critical", "RUL Healthy", "RUL Low" and "RUL Medium" change constantly across the timeline, in fact the percentages go up and down every day. This shows **a continuous reaction to the machine working over time**.

Add Life Tracks

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5. Conclusion

5.1. Research Paper Summary

This research analysed data from a Continuous Casting Machine (CCM). A CCM is an industrial system used to convert liquid steel into solid billets. The main focus of the study was the mold **sleeve**, a water-cooled copper component that initiates steel solidification. Mold sleeves degrade over time due to thermal load and mechanical friction.

The primary objective was to analyse and estimate the Remaining Useful Life (RUL) of mold sleeves. RUL represents the amount of steel (in tons) that can be processed before a sleeve must be replaced. To achieve this objective, several knowledge discovery and data mining methods were applied to extract patterns and rules from real production data.

5.2. Knowledge Discovery Overview

Formal Concept Analysis

5.3. Final Statement

The application of structured knowledge discovery methods provides a clear link between raw industrial sensor data and actionable maintenance insights. These methods support improved understanding of equipment degradation and enable more effective decision making in steel production systems.

5.4. Turnitin Similarity Check

Turnitin is a widely used academic integrity platform used to evaluate the originality of written documents. **It compares submitted texts against a large database of academic publications, web sources and previously submitted student papers**. The system generates a similarity report that highlights overlapping content and provides a similarity score.

A similarity score of approximately 10% is considered very good in most academic contexts. This level indicates that only a small portion of the text matches existing sources and these matches are typically due to standard terminology, references or commonly used technical expressions. Such a result generally reflects a high level of originality in the written work while still maintaining appropriate academic referencing standards.

More information about the platform is available at: TurnItIn

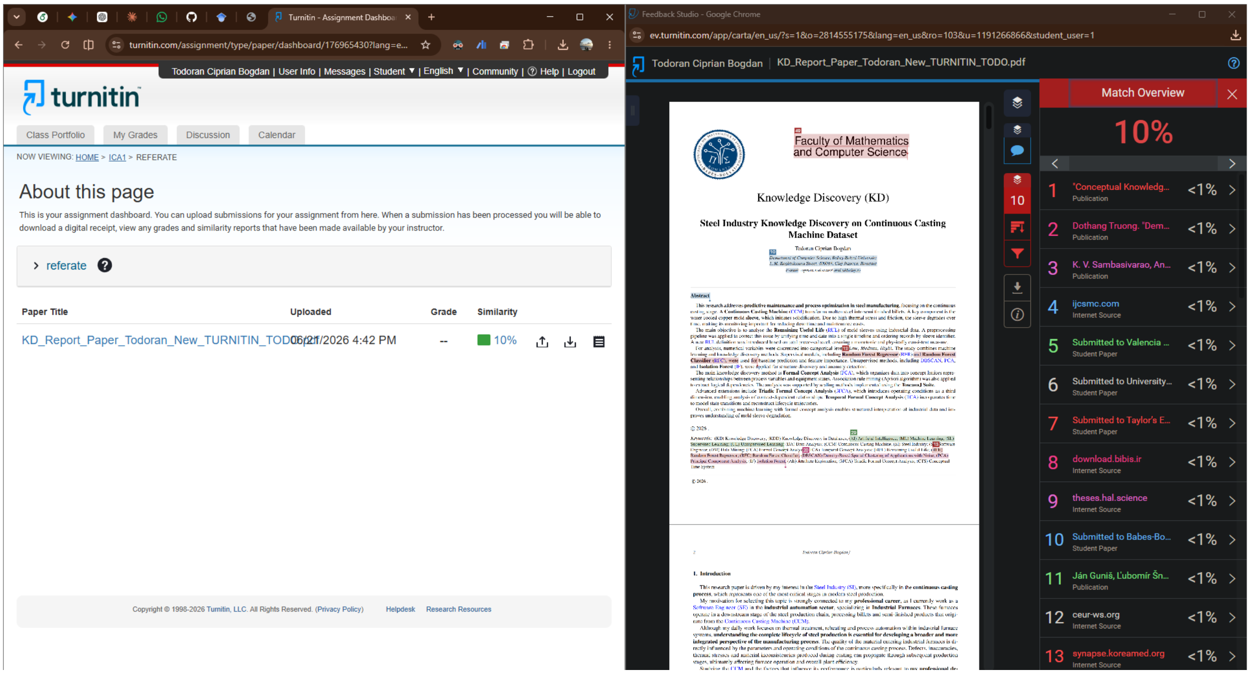

Fig. 20: TurnItIn Similarity Score Overview

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Acronyms

3FCA Triadic Formal Concept Analysis. 1, 2, 14, 15, 19

AE Attribute Exploration. 11

CCM Continuous Casting Machine. 1, 19, 21

DBSCAN Density-Based Spatial Clustering of Applications with Noise. 1

FCA Formal Concept Analysis. 1–3, 5, 7, 11, 12, 14, 20

IF Isolation Forest. 1

KD Knowledge Discovery. 3

PCA Principal Component Analysis. 1

RFC Random Forest Classifier. 1
RFR Random Forest Regressor. 1
RUL Remaining Useful Life. 1, 5, 7–9, 17, 21

TCA Temporal Concept Analysis. 1, 20